

Introduction to Data Science: Data pre-processing

Tutorial 4

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1. Consider a data set consisting of 50 observations in 5 different variables X_1, X_2, X_3, X_4 and X_5 . We use the notation $x_{i,j}$ to denote the i -th observation of X_j and we assume this data set is standardized. Assume that Z_1 and Z_2 can be defined as follows:

$$\begin{aligned}Z_1 &= \alpha_{1,1}X_1 + \alpha_{1,2}X_2 + \alpha_{1,3}X_3 + \alpha_{1,4}X_4 + \alpha_{1,5}X_5, \\Z_2 &= \alpha_{2,1}X_1 + \alpha_{2,2}X_2 + \alpha_{2,3}X_3 + \alpha_{2,4}X_4 + \alpha_{2,5}X_5.\end{aligned}$$

The vectors $\underline{\alpha}_1$ and $\underline{\alpha}_2$ are assumed to have norm 1.

- (a) True or false? Explain your answer. $\sum_{i=1}^{50} \left(\sum_{j=1}^5 \alpha_{1,j} x_{i,j} \right)^2 = \sum_{i=1}^{50} \left(\sum_{j=1}^5 \alpha_{2,j} x_{i,j} \right)^2$.
- (b) Explain why it might be a good idea to additionally assume that $\sum_{j=1}^5 \alpha_{1,j} \alpha_{2,j} = 0$.
- (c) True or false? $\sum_{j=1}^5 \alpha_{1,j}^2 + \sum_{j=1}^5 \alpha_{2,j}^2 = 1$.
- (d) True or false? $\text{Var}[Z_1] = 1$.

Which of these statements is always correct?

Solution: Question a: Note that since the data is standardized, the components Z_1 and Z_2 are also having mean zero. Then $\sum_{i=1}^{50} \left(\sum_{j=1}^5 \alpha_{1,j} x_{i,j} \right)^2$ is proportional to the variance of Z_1 whereas $\sum_{i=1}^{50} \left(\sum_{j=1}^5 \alpha_{2,j} x_{i,j} \right)^2$ is proportional to the variance of Z_2 . The variances of Z_1 and Z_2 are in general not the same and therefore this equality is not true.

Question b. This condition states that the vectors $\underline{\alpha}_1$ and $\underline{\alpha}_2$ are orthogonal. It means that we transform the variables $X_1, \dots, X_5$ into two new variables Z_1 and Z_2 which are orthogonal.

Question c. The vectors $\underline{\alpha}_1$ and $\underline{\alpha}_2$ are having norm 1, which implies that $\sum_{j=1}^5 \alpha_{1,j}^2 = 1$ and $\sum_{j=1}^5 \alpha_{2,j}^2 = 1$. Therefore, this statement is not correct.

Question d. False.

2. Consider a data set with 1000 observations. Each observation contains 10 different variables with mean zero and equal variance. We perform a principal component analysis. We determine

the scores $z_1, z_2, \dots, z_{1000}$ for each observation using the first principal component as follows:

$$z_i = \sum_{j=1}^{10} \alpha_{1,j} x_{i,j}.$$

The total variance of the data set is equal to 5 and you are given $\sum_{i=1}^{1000} z_i^2 = 1253$. What is proportion of the variance that is explained by the first principal component?

- A. Smaller than 10%
- B. Between 10% and 20%.
- C. Between 20% and 30%.
- D. Between 30% and 40%.

Solution: The variance of the first principal component can be determined as follows:

$$\text{Var}[Z_1] = \frac{1}{n-1} \sum_{i=1}^{1000} z_i^2 = \frac{1253}{999} = 1.25.$$

The variance explained by the first principal component is given by

$$\text{Variance explained} = \frac{1.25}{5} = 0.25.$$

Therefore, the correct answer is C.

3. The Motor-Trend-Car-Road dataset consists of data that was extracted from the 1974 Motor Trend US magazine, and comprises fuel consumption and 10 aspects of automobile design and performance for 32 automobiles (1973-74 models). There are 32 observations on 11 numeric variables:

- 1 mpg Miles/(US) gallon: more powerful and heavier cars tend to consume more fuel
- 2 cyl Number of cylinders: more powerful cars often have more cylinders
- 3 disp Displacement (cu.in.): the combined volume of the engine's cylinders
- 4 hp Gross horsepower: this is a measure of the power generated by the car
- 5 drat Rear axle ratio: higher values will decrease fuel efficiency
- 6 wt Weight (1000 lbs)
- 7 qsec 1/4 mile time: the cars speed and acceleration
- 8 vs Engine block: 0 = V-shaped or 1 = straight
- 9 am Transmission: 0 = automatic or 1 = manual
- 10 gear Number of forward gears: sports cars tend to have more gears

11 carb Number of carburetors: associated with more powerful engines

Because PCA works best with numerical data, the two categorical variables (vs and am) are excluded so we are left with a matrix of 9 columns and 32 rows. The data is standardized before PCA is applied and gives the following result:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9
Standard deviation	2.3782	1.4429	0.71008	0.51481	0.42797	0.35184	0.32413	0.2419	0.14896
Proportion of Variance	0.6284	0.2313	?	0.02945	0.02035	0.01375	0.01167	0.0065	0.00247

- What is the relation between the first row (Stand. dev.) and the eigenvalues λ_i ?
- Determine the Proportion of Variance for PC3.
- How many components should be retained according to the proportion of variance explained criterion at level 95%?

Solution: Question a: Since λ_i denotes the variance of the i th PC, Y_i say, the squared value of Standard deviation equals λ_i .

Question b: Using $\lambda_3/9 = 0.71008^2/9 = 0.05602$.

Question c: We can obtain the cumulative proportion

	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9
Standard deviation	2.3782	1.4429	0.71008	0.51481	0.42797	0.35184	0.32413	0.2419	0.14896
Proportion of Variance	0.6284	0.2313	0.05602	0.02945	0.02035	0.01375	0.01167	0.0065	0.00247
Cumulative Proportion	0.6284	0.8598	0.91581	0.94525	0.96560	0.97936	0.99103	0.9975	1.00000

We keep 5 components

4. Let

$$X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim N \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, R \right)$$

where

$$R = \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}.$$

Assume that the distributions X_1 and X_2 are standardized and $r > 0$.

- Determine the eigenvectors $\underline{\alpha}_1$ and $\underline{\alpha}_2$ of the covariance matrix R .
- The results from the previous question imply that the principal components of a bivariate normal distribution are independent of the correlation coefficient r . Explain this intuitively.
- What is the effect of the correlation r when performing a principal component analysis on a bivariate normal data set.

Solution: Question a: To find the eigenvalues, we need to solve the following equation with respect to λ :

$$0 = \det(\lambda I_2 - R) = \det \begin{pmatrix} \lambda - 1 & -r \\ -r & \lambda - 1 \end{pmatrix} = (\lambda - 1)^2 - r^2,$$

which implies that

$$(\lambda - 1) = \pm r;$$

or that the eigenvalues λ are $1 \pm r$. The eigenvector $\underline{\alpha} = (\alpha_1, \alpha_2)$ is given by

$$(R - \lambda I)\underline{\alpha}' = \underline{0}$$

When $\lambda_1 = 1 + r$, we get

$$\begin{aligned} \alpha_{1,1} &= \alpha_{1,2} \\ \alpha_{1,1}^2 + \alpha_{1,2}^2 &= 1 \end{aligned}$$

Thus $\alpha_{1,1} = \alpha_{1,2} = 1/\sqrt{2}$. Similarly when $\lambda_2 = 1 - r$, $\alpha_{2,1} = 1/\sqrt{2}$ and $\alpha_{2,2} = -1/\sqrt{2}$.

Question b: The observations from a bivariate normal distribution with correlation r are always centered around the line $y = x$. Therefore, the principal components will be based on the orthogonal lines $y = x$ and $y = -x$.

Question c: The correlation r determines the proportion of the variance that is explained by each of the principal components. For the first principal component, the variance explained is given by:

$$\text{Variance explained by the first PC} = \frac{\lambda_1}{\text{Var}[X_1] + \text{Var}[X_2]} = \frac{1 + r}{2}.$$

We find that the first principal component explains more variance when r is becoming larger. This is intuitively clear, since a larger value for the correlation r implies that the observations are ‘closer’ to the line $y = x$, which is exactly the first principal component.

5. Consider a data set with car information.

Variable	PC 1	PC2
gas mileage (MPG)	0.3	0.02
Weight	-0.36	0.18
Length	-0.27	0.42
Engine	-0.34	0.02
Horsepower	-0.32	-0.29
Price	-0.26	-0.47

Give an interpretation of the first and second principal component.

Solution: The first principal component is influenced equally by all the variables except for gas mileage. Moreover, gas mileage has the opposite sign, but same magnitude. Note, however, that a large number for gas mileage implies a car that consumes a lot of fuel. Therefore, larger cars will have lower gas mileage. Therefore, the first principal component corresponds with the general size of the car. More expensive, large cars on one end of the spectrum to smaller, fuel efficient cars on the other end of the spectrum.

The second principal component only very weakly depends on the variables gas mileage and engine. The effect of horsepower is the opposite than the effects of weight and length. Therefore, we can think of the second principal component as the variable that separates large family cars that are relatively cheap from the expensive sports cars (with lots of horsepower).

6. Consider the random variables $\underline{X} = (X_1, X_2, \dots, X_p)$ with covariance matrix denoted by Σ . Define the variable Z_1 as a linear combination of the variables $\underline{X}$:

$$Z_1 = \alpha_{1,1}X_1 + \alpha_{1,2}X_2 + \dots + \alpha_{1,p}X_p.$$

Show that

$$\text{Var}[Z_1] = \underline{\alpha}_1 \Sigma \underline{\alpha}_1',$$

where $\underline{\alpha}_1 = (\alpha_{1,1}, \alpha_{1,2}, \dots, \alpha_{1,p})$.

Solution: We can write the variance of Z_1 as follows

$$\begin{aligned} \text{Var}[Z_1] &= \sum_{i=1}^p \sum_{j=1}^p \alpha_{1,i} \alpha_{1,j} \text{Cov}[X_i, X_j] \\ &= \sum_{i=1}^p \alpha_{1,i} \left(\sum_{j=1}^p \alpha_{1,j} \sigma_{i,j} \right) \\ &= (\alpha_{1,1}, \alpha_{1,2}, \dots, \alpha_{1,p}) \begin{pmatrix} \sum_{j=1}^p \sigma_{1,j} \alpha_{1,j} \\ \sum_{j=1}^p \sigma_{2,j} \alpha_{1,j} \\ \vdots \\ \sum_{j=1}^p \sigma_{p,j} \alpha_{1,j} \end{pmatrix}. \end{aligned}$$

It is then easy to verify that

$$\Sigma \underline{\alpha}_1' = \begin{pmatrix} \sum_{j=1}^p \sigma_{1,j} \alpha_{1,j} \\ \sum_{j=1}^p \sigma_{2,j} \alpha_{1,j} \\ \vdots \\ \sum_{j=1}^p \sigma_{p,j} \alpha_{1,j} \end{pmatrix},$$

which ends the proof.

7. Consider the covariance matrix Σ of a n -dimensional random vector $\underline{X}$. Define $\underline{\alpha}_1 = (\alpha_{1,1}, \alpha_{1,2}, \dots, \alpha_{1,n})$ and

$$f(\underline{\alpha}_1) = \underline{\alpha}_1 \Sigma \underline{\alpha}'_1 - \lambda (\underline{\alpha}_1 \cdot \underline{\alpha}_1 - 1).$$

Show that

$$\frac{\partial f}{\partial \underline{\alpha}_1} = 2 (\Sigma \underline{\alpha}'_1 - \lambda \underline{\alpha}'_1).$$

Solution: Note that:

$$\frac{\partial f}{\partial \underline{\alpha}_1} = \begin{pmatrix} \frac{\partial f}{\partial \alpha_{1,1}} \\ \frac{\partial f}{\partial \alpha_{1,2}} \\ \vdots \\ \frac{\partial f}{\partial \alpha_{1,n}} \end{pmatrix}.$$

We therefore first calculate $\frac{\partial f}{\partial \alpha_{1,k}}$. We have that

$$\begin{aligned} \underline{\alpha}_1 \Sigma \underline{\alpha}'_1 &= \sum_{i=1}^n \alpha_{1,i} \left(\sum_{j=1}^n \alpha_{1,j} \sigma_{i,j} \right) \\ &= \sum_{i=1, i \neq k}^n \alpha_{1,i} \left(\sum_{j=1}^n \alpha_{1,j} \sigma_{i,j} \right) + \sum_{j=1}^n \alpha_{1,k} \alpha_{1,j} \sigma_{k,j} \\ &= \sum_{i=1, i \neq k}^n \alpha_{1,i} \left(\sum_{j=1}^n \alpha_{1,j} \sigma_{i,j} \right) + \sum_{j=1, j \neq k}^n \alpha_{1,k} \alpha_{1,j} \sigma_{k,j} + \alpha_{1,k}^2 \sigma_{k,k} \end{aligned}$$

Then:

$$\begin{aligned} \frac{\partial \underline{\alpha}_1 \Sigma \underline{\alpha}'_1}{\partial \alpha_{1,k}} &= \sum_{i=1, i \neq k}^n \alpha_{1,i} \sigma_{i,k} + \sum_{j=1, j \neq k}^n \alpha_{1,j} \sigma_{k,j} + 2\alpha_{1,k} \sigma_{k,k} \\ &= 2 \sum_{i=1, i \neq k}^n \alpha_{1,i} \sigma_{i,k} + 2\alpha_{1,k} \sigma_{k,k} \\ &= 2 \sum_{i=1}^n \alpha_{1,i} \sigma_{i,k}. \end{aligned}$$

Note that we can write the following:

$$\begin{aligned} \Sigma \underline{\alpha}'_1 &= \begin{pmatrix} \sigma_{1,1} & \sigma_{1,2} & \dots & \sigma_{1,n} \\ \sigma_{2,1} & \sigma_{2,2} & \dots & \sigma_{2,n} \\ \vdots & \vdots & \vdots & \vdots \\ \sigma_{n,1} & \sigma_{n,2} & \dots & \sigma_{n,n} \end{pmatrix} \begin{pmatrix} \alpha_{1,1} \\ \alpha_{1,2} \\ \vdots \\ \alpha_{1,n} \end{pmatrix} \\ &= \begin{pmatrix} \sum_{i=1}^n \alpha_{1,i} \sigma_{1,i} \\ \sum_{i=1}^n \alpha_{1,i} \sigma_{2,i} \\ \vdots \\ \sum_{i=1}^n \alpha_{1,i} \sigma_{n,i} \end{pmatrix} \end{aligned}$$

If we use that $\sigma_{j,k} = \sigma_{k,j}$, we can write

$$\frac{\partial \underline{\alpha}_1 \Sigma \underline{\alpha}'_1}{\partial \alpha_{1,k}} = 2 \sum_{i=1}^n \alpha_{1,i} \sigma_{1,k},$$

and therefore

$$\frac{\partial \underline{\alpha}_1 \Sigma \underline{\alpha}'_1}{\partial \underline{\alpha}_1} = 2 \Sigma \underline{\alpha}'_1.$$

We also have that

$$\frac{\partial \lambda (\underline{\alpha}_1 \cdot \underline{\alpha}'_1 - 1)}{\partial \alpha_{1,k}} = 2 \lambda \alpha_{1,k},$$

which implies that

$$\frac{\partial \lambda (\underline{\alpha}_1 \cdot \underline{\alpha}'_1 - 1)}{\partial \underline{\alpha}_1} = 2 \lambda \underline{\alpha}_1,$$

Combining these formulas proves the result.

Exam question

1. Consider a data set with two variables, denoted by X_1 and X_2 . You are also given the following information

$$\mathbb{E}[X_1] = 178.5, \quad \mathbb{E}[X_2] = 76.5, \quad \text{Var}[X_1] = 275.67, \quad \text{Var}[X_2] = 126.67.$$

Moreover, you are also given that

$$\text{Var}[X_1 + X_2] = 714.$$

- (a) (4 points) Determine the first principal component. You do not need to standardize or scale the data.
- (b) (1 point) Determine the proportion of the total variance explained by the first principal component.
- (c) (2 points) Denote by (β_1, β_2) the second principal component. Determine the variance of the second principal component.

Observation	X_1	X_2
1	160	70
2	180	81
3	200	90
4	174	65

Table 1: Data set with four observations.